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Flatness Filtering for Honeyword Generation

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Abstract

Honeyword schemes defend password databases against offline dictionary attacks by storing decoy passwords, or sweetwords, alongside the real one, so that a stolen hash file cannot be used without risking detection. Their security rests on flatness: the real password must be indistinguishable from its decoys. Dani, McCulloh, and Saxena [DMS24] have shown that a deep-learning classifier can pick the real password out of a sweetword set, which breaks flatness for the honeyword generators in current use.

This thesis reuses that classifier as a defence. A CNN flatness oracle scores each candidate sweetword set and rejects the ones in which the real password stands out, regenerating until a set passes or a regeneration cap is reached. I benchmark ten generation methods on the RockYou corpus: rule-based chaffing, HoneyGen variants, and a self-trained GAN, at training sizes of 2000, 5000, and 10000. Each is evaluated against three attackers, a HoneyGen-tuned CNN, a mixed-corpus CNN, and a frequency ranker. The metric throughout is the attacker’s success rate at guess budget t (ASR@ t), the fraction of accounts compromised within t guesses against the $k = 20$ sweetwords of an account. Random guessing gives t/k .

The unfiltered generators are easy for the CNN to break. At $k = 20$ they reach ASR@1 of 0.28–0.35, ASR@5 of 0.68–0.76, and ASR@10 of 0.86–0.91, against random baselines of 0.05, 0.25, and 0.50. Ratio-based filtering is the strongest defence that remains deployable, cutting those figures to 0.05–0.07, 0.29–0.34, and 0.58–0.63, and leaving the real password near the middle of the attacker’s ranking. Reaching that point means treating the acceptance rate as a deployment requirement rather than a diagnostic.

The GAN deserves a warning. Its reported ASR@5 is 0.87, which looks nothing like the near-zero value a naive CNN attacker measures. The reason is that the real password sits at the bottom of the ranking ($F_{\text{rank}} \approx 0.92$), so an attacker who inverts the ranking recovers it almost every time. A mixed-corpus attacker pushes the same sets close to 1. What the GAN produces is inverted distinguishability rather than flatness.

Keywords: Honeywords, passwords, authentication, flatness, deep learning.

Declaration on the Use of AI Tools

In accordance with the University of Genoa guidelines on the use of artificial intelligence (Linee guida IA, Art. 3 and Art. 10), I declare the following.

Claude (Anthropic) was used as a writing assistant while I prepared this document, in three ways. After I had read the relevant literature myself, it helped me organise and summarise that material and phrase the comparisons in Chapter 3. It also helped me analyse my experimental results and generating the L^AT_EX source of the tables and figures from my data. in addition, drafted and revised prose, which is the most substantial of the three. Sections of Chapter 4 and Chapter 6 along with the abstract, were drafted with myself and the assistant working together, and then reviewed and edited by me. Sections of Chapter 5 were drafted with me and reviewed by its assistant, and then edited by me.

The research reported here is my own. I designed the filtering method, implemented the pipeline, trained all the models, ran every experiment, and produced all the data, curves and tables. Every number in this thesis comes from my own runs. I read the assistant's output critically and corrected it where it was wrong.

No personal data were supplied to the tool, and the password corpora used in this work were never uploaded to it. I have verified the technical content, the numerical values and the citations, and I take full responsibility for this thesis.

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Chapter 1

Introduction

Password authentication remains the dominant credential mechanism for consumer web services, in spite of decades of proposals for alternatives. It persists because it is easy to deploy, familiar to users, and costs almost nothing per account. The price is a long history of database breaches that expose hundreds of millions of hashed credentials to offline guessing. Once a hash file has left the server, the guessing rate is bounded only by the attacker’s compute, and any weakness in the underlying password distribution turns directly into recovered plaintexts.

Juels and Rivest [JR13] proposed *honeywords* as a detection layer over this failure mode. For each account the authentication server stores a *sweetword set* of k candidate hashes, one of which is the real password and the other $k - 1$ of which are decoys, together with a separate index held by a hardened *HoneyChecker* that records which position is real. A successful login with a decoy signals that the hash file has been read offline and that the attacker has begun trying candidates. The scheme does not prevent guessing. It converts silent database compromise into an event the server can detect and act on.

Its security rests on *flatness*: given the sweetword set, an attacker who has read the hash file must not be able to identify the real password with probability much better than $1/k$. Under perfect flatness all k candidates look alike to the attacker, and every login attempt against a decoy raises the alarm with the same probability. Flatness is what turns honeywords from a labelling exercise into a detection guarantee.

1.1 Motivation

In most of the honeyword literature, flatness is measured against attackers that no longer represent the state of the art. Traditional evaluations rank sweetwords using frequency

counts, probabilistic context-free grammars, or Markov models, and honeywords are declared flat when the real password does not stand out under those rankers. Dani, McCulloh, and Saxena [DMS24] showed that a modestly sized character-level convolutional neural network, trained as a binary classifier, defeats this notion on every scheme it was tested against, including the representation-learning generators of Dionysiou, Vassiliades, and Athanasopoulos [DVA21]. Under a learned discriminator the attacker success rate runs well above $1/k$, sometimes by an order of magnitude, and sweetwords that pass heuristic evaluation turn out to be individually plausible but collectively distinguishable.

Two consequences motivate the rest of the thesis. Current honeyword generators have almost never been benchmarked against a neural attacker, so how badly they fail once one enters the threat model is simply unknown. And no construction in the literature restores flatness against such an attacker without redesigning the generator, which would leave existing deployments unprotected. Both need resolving if honeyword-based breach detection is to remain credible.

1.2 Approach

We take the discriminator that Dani, McCulloh, and Saxena [DMS24] introduced as an attack tool and reuse it, unmodified in architecture, as a *defensive oracle* during sweetword construction. A candidate sweetword set is produced by any of the standard mechanisms, scored by the oracle, and accepted or discarded depending on whether the induced score distribution over the set satisfies a small set of measurable flatness conditions. Sets that fail are regenerated up to a bounded budget. The classifier that would otherwise identify the real password becomes the gatekeeper that keeps any set with an identifiable real password from being stored.

The construction is deliberately generator-agnostic. We apply it as a post-processing layer over ten honeyword generation methods, spanning rule-based tweaking, representation-learning approaches derived from Dionysiou, Vassiliades, and Athanasopoulos [DVA21], and a self-trained GAN. Each method is trained once at a fixed training-set size and evaluated on a common test set of sweetword sets, so comparisons across generators are made on the same footing. The primary evaluation uses a CNN attacker trained on HoneyGen; robustness is then checked against two further attacker classes, a mixed-corpus CNN trained on several decoy families and a frequency ranker, which together separate genuine flatness from single-attacker artifacts.

The primary security metric is the attacker’s success rate at guess budget five, ASR@5: the fraction of accounts an attacker compromises when allowed five guesses among the $k = 20$ sweetwords of an account. We report the full curve ASR@ t for $t = 1, \dots, 20$ but summarise on $t = 5$, because that matches the honeyword threat model. After cracking

a stolen hash file offline, an attacker still has to submit a recovered sweetword to the live service; submitting a honeyword trips the honeychecker, and most online services lock an account after a small number of failed attempts, five being a common threshold [DMS24]. ASR@5 thus measures what an attacker can achieve inside one realistic lockout window. The extremes make poor headlines: ASR@1 reflects only the single best guess, and ASR@20 equals 1.0 by construction once all $k = 20$ sweetwords have been tried. For a perfectly flat set the attacker does no better than random, giving $\text{ASR@5} = 5/20 = 0.25$.

1.3 Contributions

The thesis makes the following contributions.

1. **Attacker-relative set-level flatness.** We define flatness as a property of the probability distribution that a learned attacker induces over a sweetword set, and characterise it by four quantities computed from that distribution: rank-based flatness, normalized entropy, top and real gaps, and an ε -flatness bound on the largest renormalized score in the set. These replace the frequency-based flatness notions of the earlier literature and make the property measurable under any binary discriminator.
2. **Hybrid flatness filtering.** We propose a two-stage rejection-sampling algorithm that combines candidate-level pre-screening with set-level acceptance. Pre-screening removes individually distinguishable decoys before a set is assembled; set-level acceptance discards sets whose induced distribution violates any of the flatness conditions. The algorithm works as a post-processing layer and requires no change to the underlying generator and no retraining of the attacker.
3. **Benchmark of ten honeyword generators against a neural attacker.** We evaluate Chaffing-by-Tweaking, Chaffing-with-Password-Model, Chaffing-with-Hybrid-Model, HoneyGen Baseline, HoneyGen Set-Filter, HoneyGen Prescreen-Add, HoneyGen Prescreen-Ratio, HoneyGen Hybrid-Add, HoneyGen Hybrid-Ratio, and a self-trained GAN, on the RockYou corpus with $k = 20$. Reporting the same attacker success rate under a common protocol lets us separate real flatness gains from distributional artifacts of the generator, which turns out to matter for the GAN and which we discuss in Chapter 5.
4. **A defensive reinterpretation of the neural-attacker paradigm.** The thesis shows that a discriminator of the type Dani, McCulloh, and Saxena [DMS24] used as an attack can be repurposed, without architectural change, into the enforcement mechanism of a flatness guarantee. What defeats current honeyword schemes at test time becomes what protects them at storage time.

1.4 Scope and Assumptions

The guarantees are model-relative. Filtering is performed against one specific attacker, the character-level CNN described in Chapter 4, and the flatness conditions constrain that attacker’s induced distribution over the sweetword set. A different attacker, with a different architecture or with access to side information beyond the sweetword strings, may recover distinguishability. Chapter 6 returns to this when discussing the limits of the framework.

The evaluation is restricted to leaked passwords from the RockYou corpus and to sweetword sets of size $k = 20$. Only the password string itself is used as input by the generators and the attackers. Chapter 5 reports how far the conclusions carry across attacker classes, from the HoneyGen-tuned CNN to a mixed-corpus CNN and a frequency ranker; extending the benchmark to a second corpus such as *darkc0de* is left as future work (Chapter 6).

The rest of the thesis proceeds as follows. Chapter 2 sets out the honeyword primitive, the HoneyChecker architecture, and the character-level convolutional networks that both the attack and the defence rely on. Chapter 3 reviews the generators we benchmark against and situates Dani, McCulloh, and Saxena [DMS24] within the wider line of neural password modelling. Chapter 4 develops the flatness formalism, the CNN training procedure, the ten generators, the two-stage filtering algorithm, and the evaluation protocol. Chapter 5 reports the ASR curves across the ten generators, isolates the effect of filtering on both security and the flatness criteria, weighs it against the acceptance rate as a deployability requirement, checks robustness across attacker classes, and discusses a distributional artifact in the GAN results that limits how far its ASR should be trusted. Chapter 6 draws the argument together and lists the open questions the framework leaves behind.

Chapter 2

Background

This chapter collects the definitions and mechanisms that the rest of the thesis takes as given. Section 2.1 describes how offline password guessing works and what a breach exposes. Section 2.2 states the honeyword primitive introduced by Juels and Rivest [JR13] and the HoneyChecker architecture that supports it. Section 2.3 formalizes the classical notion of flatness relative to a fixed attacker class. Section 2.4 introduces the character-level convolutional networks that Chapter 3 and Chapter 4 both build on, and Section 2.5 covers the FastText subword embeddings used by the HoneyGen family of generators.

Every concept is presented at the level required to follow the argument in later chapters. The formal set-level flatness definition that this thesis introduces is deferred to Section 4.2, since it is part of the contribution rather than of the background.

2.1 Password Guessing After a Breach

Password authentication systems store a transformation of the user’s credential rather than the credential itself. In modern deployments this transformation is a salted, slow cryptographic hash such as bcrypt, scrypt, or Argon2. When a user submits a password w at login, the server computes the hash and compares it to the stored value; a match authenticates the session.

Two attack models are relevant here. In the *online* model the attacker submits guesses through the authentication interface, and the server can rate-limit or lock accounts after a small number of failed attempts. In the *offline* model the attacker has obtained a copy of the password file and can test guesses locally at the rate their hardware allows. Modern GPU rigs push this rate high enough that the security of the deployment is determined almost entirely by the password’s position in the guessing order used by the attacker.

The guessing order itself has been the subject of a long research line. Narayanan and Shmatikov [NS05] showed that Markov models of character sequences accelerate cracking substantially over dictionary attacks. Weir et al. [Wei+09] extracted probabilistic context-free grammars from leaked corpora and produced guesses in decreasing order of estimated probability. Melicher et al. [Mel+16] used neural language models over character sequences to reach guessing rates that outperform both Markov and PCFG models on realistic budgets. All three lines share the same interface: given a candidate password, produce a score or a rank that reflects how likely it is to appear in a real distribution.

The consequence for defenders is that a database compromise is silent. Nothing distinguishes an offline guessing campaign from a period in which no attacker has the hash file at all, until the attacker’s recovered credentials are used against a live login. Honeywords, the subject of the next section, are designed to break that silence.

2.2 The Honeyword Primitive

2.2.1 Sweetword Storage

Juels and Rivest [JR13] propose the following modification to the password file. For each user account the server stores not a single hash but a set of k hashes, one of which corresponds to the real password and the other $k - 1$ to *honeywords*: decoy strings generated to be difficult to distinguish from the real password. We call the set of k candidates the *sweetword set*, and any element of it a *sweetword*. The honeywords are not stored in clear anywhere; only their hashes appear in the password file.

The order of the sweetwords in the file is either randomized per user or made independent of the identity of the real password. The information about which position corresponds to the real password is stored elsewhere, in a component we describe next.

2.2.2 The HoneyChecker

The scheme separates two pieces of information that were previously combined in a single password file: the credential material (the k hashes) and the identity of the correct one (an index in $\{1, \dots, k\}$). The credential material stays on the main authentication server, and the index is held by a second, hardened component called the *HoneyChecker*. The HoneyChecker exposes a narrow interface: given a user identifier and an index, it returns whether the index is the correct one, and it raises an alarm if the index is not.

An honest login proceeds as follows. The user submits their password. The authentication server hashes it and looks for a match among the k stored sweetword hashes. If a match is

found at index j , the server queries the HoneyChecker with that index. The HoneyChecker returns *accept* when j is the correct index, and the login succeeds. This is indistinguishable to the user from an ordinary password check.

A honeyword-triggered login proceeds differently. The attacker, who has read the password file offline and recovered one of the k hashes, attempts a login using the recovered plaintext. The authentication server finds a match at some index j' and queries the HoneyChecker. If j' is not the correct index, the HoneyChecker returns *reject* and, in addition, raises an alarm. The alarm is what makes the primitive a breach detector: a decoy login is observable in real time, whereas a successful compromise otherwise is not.

The distributed trust assumption is that a single adversary does not compromise both the password file and the HoneyChecker. Reading only the password file leaves the attacker with k indistinguishable hashes; compromising only the HoneyChecker gives away indices with no credential material to attach them to. This assumption is what makes the scheme meaningful; Juels and Rivest [JR13] and the subsequent survey by Chakraborty et al. [Cha+23] discuss its practical implications at length, and we take it as given throughout the thesis.

2.2.3 Alarms and Response

Once the HoneyChecker raises an alarm, the response is a policy decision outside the scope of this thesis. Depending on the deployment, the response can lock the affected account, force a password reset for a class of users, or trigger a wider incident investigation. The relevant property here is that the alarm is a signal that the server did not have before the primitive was in place.

2.3 Classical Flatness

2.3.1 The Attacker’s Task

Given only the sweetword set $S = \{w_1, \dots, w_k\}$, the attacker must decide which element is the real password. This is a k -way distinguishing problem. If the honeywords are well constructed, the attacker’s success probability should be close to $1/k$, the rate of uniform random guessing; if the honeywords are distinguishable, the success probability is higher.

The exact success probability depends on the attacker’s knowledge and inference capabilities. Juels and Rivest’s original paper models the attacker as a probabilistic ranker: a function A that, given a candidate password, returns a probability that it comes from the

distribution of real user-chosen passwords. The attacker orders the k sweetwords by A 's output and guesses the top-ranked one.

2.3.2 $1/k$ -Flatness and ε -Flatness

Let A be an attacker and let $p_A(S)$ denote the probability that A identifies w^* on its first guess, taken over the random choice of S and the internal randomness of the generator. The scheme is called $1/k$ -flat against A if

$$p_A(S) = 1/k, \quad (2.1)$$

which is the uniform-random rate. Since perfect equality is rarely attainable, the literature relaxes the condition to ε -flatness: the scheme is ε -flat if

$$p_A(S) \leq \frac{1 + \varepsilon}{k}, \quad (2.2)$$

for some fixed $\varepsilon > 0$. The two definitions coincide as $\varepsilon \rightarrow 0$.

Both definitions are attacker-relative in principle: they depend on the class of function A used. The classical literature restricts A to the probabilistic ranker family described above. Chapter 4 extends the definition to arbitrary binary discriminators and formalizes it at the level of the induced distribution over the set; that extension is one of the contributions of this thesis.

2.3.3 Related Security Properties

Chakraborty et al. [Cha+23] organize the honeyword security literature around three properties: flatness, denial-of-service (DoS) resistance, and multiple-system vulnerability (MSV) resistance. DoS resistance controls the rate at which a legitimate but confused user can be locked out by accidentally submitting a honeyword. MSV resistance controls the extent to which learning the sweetwords for one user helps an attacker for other users. This thesis addresses only flatness. The DoS and MSV properties are largely orthogonal to the filtering construction we develop, and we return to them briefly in Chapter 6.

2.4 Character-Level Convolutional Classifiers

Character-level convolutional networks are the standard architecture for both attack and defense in this thesis. The design is a small adaptation of the text classifier introduced by Kim [Kim14] and the character-level variants of Zhang, Zhao, and LeCun [ZZL15].

A password w of at most L characters is right-padded to length L and each character is mapped to an embedding vector of dimension d through a lookup table. The resulting $L \times d$ matrix is treated as a one-dimensional signal with d input channels. One or more one-dimensional convolutional layers extract features from local n -character windows: each convolution with kernel width m applies a learned filter to every window of m consecutive positions and produces a new feature map. A ReLU activation follows each convolution. A max-pooling step reduces the length dimension, and a fully connected layer projects the pooled features to the number of output classes. A final softmax converts the logits to class probabilities.

For the binary classifier used here, the two output classes are *real* and *honeyword*, and the quantity of interest is $P(\text{real} \mid w)$, the probability the network assigns to the real class for the input password. This score is the raw material for every flatness measurement in the thesis. The precise architecture and training procedure are specified in Section 4.4.

The property that makes this class of network well suited to password strings is that its filters operate on local character windows. This matches the empirical observation that real user-chosen passwords concentrate distinguishing structure at the level of short substrings (digit sequences, common suffixes, case patterns), which the convolutional layers pick up efficiently without requiring token-level segmentation.

2.5 FastText Password Embeddings

The HoneyGen generator family described in Section 4.5 and reviewed in Chapter 3 uses FastText embeddings as its representation of the password space. FastText, introduced by Bojanowski et al. [Boj+17], extends the skipgram word representation of Mikolov et al. [Mik+13] by representing each token as the sum of embeddings for its character n -grams. A password like `qwerty2020` is represented not only by an embedding for the string as a whole but by the sum of embeddings for all its character n -grams of length in a configured range.

Two properties of this representation matter for honeyword generation. First, the model produces embeddings for out-of-vocabulary strings, which is essential because a generator is asked to produce passwords it has never seen. Second, the nearest neighbors of a password in embedding space tend to share sub-word structure: a password ending in `2020` is close to passwords ending in `2021`, `2019`, and so on. This makes the embedding a source of plausible perturbations of a real password without requiring hand-written rules.

The HoneyGen family uses a skipgram FastText model trained on the oracle corpus, with the smallest possible `minCount` and a minimum n -gram length of two. Section 4.5 gives the exact configuration used in the experiments.

Chapter 3

Related Work

The literature on honeyword-based authentication runs along two threads. The first is the design of *generators* that produce honeywords meant to be hard to tell apart from real passwords. The second, more recent, is the design of *distinguishers* that try to break flatness under stronger attacker models than generators are usually tested against. Both feed into this thesis: the ten generators of Section 4.5 come from the first thread, the CNN of Section 4.4 from the second. This chapter reviews both, and Section 3.7 describes how the filtering construction of Section 4.6 sits between them.

3.1 Heuristic Honeyword Generators

The earliest generators are the four *chaffing* constructions proposed alongside the primitive by Juels and Rivest [JR13]: *chaffing-by-tweaking*, *chaffing-by-tail-tweaking*, *chaffing-with-a-password-model*, and *chaffing-with-tough-nuts*. Chaffing-by-tweaking applies character-class substitutions to the real password to produce visually similar decoys; chaffing-by-tail-tweaking restricts those edits to a short suffix. Chaffing-with-a-password-model draws candidates from an external password model such as a PCFG or a Markov chain, so the decoys look like plausible passwords in the aggregate rather than like edits of the real one. Chaffing-with-tough-nuts inserts random, high-entropy strings that no attacker would submit at login and that therefore work as guaranteed alarms.

These were evaluated against attackers that scored candidates independently, typically the PCFG rankers of Weir et al. [Wei+09] and the Markov rankers of Narayanan and Shmatikov [NS05]. Under that evaluation model, carefully tuned chaffing schemes report flatness close to $1/k$, which is what motivated the follow-up literature. Two of the constructions, chaffing-by-tweaking and chaffing-with-a-password-model, appear in our benchmark (Section 4.5)

as reference points against the more sophisticated generators below, together with a hybrid chaffing method that mixes the two and that we introduce for comparison.

The main criticism of the heuristic generators is *per-password plausibility*. Each decoy is designed to look like a plausible password on its own, with no constraint on how the decoys look *as a set*. A generator can produce $k - 1$ plausible but distributionally clustered decoys that leave the real password stranded, and the attacker need only notice the cluster to spot the outlier. Later generators, and the filter of Section 4.6, are built to address this.

3.2 Representation-Learning Approaches: HoneyGen

Dionysiou, Vassiliades, and Athanasopoulos [DVA21] propose *HoneyGen*, which combines a FastText password model with rule-based tweaking. The password model produces nearest-neighbour candidates for the real password in the FastText embedding space (Section 2.5), while a tweaking module produces further candidates by character-class substitution. The two streams are merged, pruned with a frequency-based filter that removes overly common candidates, and the top $k - 1$ are kept as honeywords.

The original paper evaluates against frequency-based and PCFG-based attackers, and HoneyGen improves on the chaffing baselines. Against that class of attacker the conclusion is uncontroversial: representation learning captures more of the password distribution than rules or PCFGs alone, and the resulting decoys are harder to separate under a ranker of the kind those attackers use.

Our position on HoneyGen is specific. We reuse the generator unmodified as the candidate source for six of the ten methods in the benchmark (HoneyGen Baseline, Set-Filter, Prescreen-Add, Prescreen-Ratio, Hybrid-Add, Hybrid-Ratio). What we do not reuse is the paper’s evaluation: our attacker is the CNN of Section 2.4 rather than a frequency ranker, and the flatness scores we measure under it are far worse than the ones originally reported (Chapter 5). The filtered variants are how we recover flatness against that stronger attacker without touching the generator.

3.3 Large-Language-Model Approaches

More recent work uses large language models as the candidate source. The chunk-level construction of Yu and Vargas Martin [YV23] has drawn particular attention: a password is disassembled into semantic chunks that a pre-trained language model perturbs one at a time, and candidate sets are drawn with Metropolis–Hastings sampling, an accept-or-reject scheme that steers candidates toward regions the model assigns a high real-password

probability. Locality-sensitive hashing, which maps similar strings to the same bucket, then drops near-duplicates from the pool. These are among the most linguistically sophisticated generators the honeyword literature has produced, and they report clear gains in semantic plausibility over both HoneyGen and GAN-based methods.

The evaluation in this line, though, follows the same statistical plausibility paradigm as the earlier work. Flatness is measured against ranker-style attackers, and the reported metrics are distributional distances rather than the success rate of a neural distinguisher trained to identify the real password. The gap this leaves is the one Dani, McCulloh, and Saxena [DMS24] demonstrated for HoneyGen: a generator can be indistinguishable under one attacker class and break under a stronger one. We do not evaluate an LLM-based generator here, since our focus is the reuse of the neural distinguisher as a defensive oracle rather than the generation side. The construction of Section 4.6 is agnostic to where candidates come from and would apply to an LLM-based generator unchanged; Chapter 6 returns to this.

3.4 GAN-Based Honeyword Generators

Hitaj et al. [Hit+19] adapt the generative adversarial network of Goodfellow et al. [Goo+14] to password modelling in the attack setting: the goal is guess sequences that recover more of a leaked corpus than PCFG or Markov models manage at the same budget. The adaptation is to train the generator against a discriminator that tries to separate generated strings from real leaked ones, with both networks working on character sequences.

GANs have already been studied as a honeyword source. Fauzi, Yang, and Martiri [FYM20] analyse a game in which the defender uses the generator of a PassGAN model to produce honeywords while the attacker uses its discriminator to pick out the real password, and assess how feasible PassGAN is as a generation strategy. We include a GAN in the same spirit as one of the benchmark generators: a model that has learned the password distribution should, in principle, produce decoys that look real. Ours is a character-level generator trained from scratch on the same corpus as the other generators (a *self-trained* GAN), and we sample decoys from it directly. GANs are prone to *mode collapse*, where the generator emits only a narrow set of outputs instead of covering the variety of the training distribution; for passwords that shows up as the same few common strings appearing over and over. To get enough distinct decoys anyway, we oversample the generator, drawing roughly five times as many candidates as a set needs, and deduplicate the pool before assembling each sweetword set. This GAN is one of the ten generators evaluated in the benchmark (Chapter 4).

The theoretical case for a GAN as a honeyword generator is subtle, and our results show it does not hold in practice. A generator that had truly converged to the real password dis-

tribution would produce indistinguishable decoys almost by definition. But real password distributions have long low-probability tails that GANs are known to underrepresent, so the decoys cluster in the high-probability region and the real password, itself often a low-probability string, is left looking anomalous. Chapter 5 shows exactly that: under the CNN attacker the GAN’s real password is pushed to the bottom of the ranking ($F_{\text{rank}} \approx 0.92$), so its apparently low attack success is an artifact of that inversion rather than genuine flatness, and a mixed-corpus attacker exposes it. The GAN’s flatness numbers have to be read with care rather than taken at face value.

3.5 Neural Distinguishers as Attackers: PassFilter

The work most directly relevant to this thesis is PassFilter [DMS24]. It trains a character-level convolutional neural network as a binary classifier that predicts, given a candidate password, whether it is real or a honeyword. The network is small by modern standards, two convolutional layers over character embeddings followed by a fully connected layer, and it is trained on labelled pairs of real passwords and honeywords produced by existing generators.

The paper’s main result is that this classifier violates flatness on every honeyword scheme it tests, including HoneyGen and its variants, across multiple leaked corpora. The attacker success rate runs well above $1/k$ throughout, sometimes by an order of magnitude, and the ranking the classifier induces concentrates on the real password with a probability the classical flatness definition rules out.

Using a learned discriminator as the attacker is not unique to PassFilter. As noted in Section 3.4, Fauzi, Yang, and Martiri [FYM20] already turn the discriminator of a PassGAN model against its own generator’s honeywords. PassFilter reads as the general form of the same idea: a classifier trained to separate real passwords from honeywords whatever generator produced them. Both establish the neural distinguisher as the attacker class honeyword schemes must withstand. This thesis takes that distinguisher and moves it to the defensive side.

Our reading of the PassFilter result is that the honeyword literature has been evaluating flatness against too weak an attacker class. The constructive response we develop is to admit the same class of classifier into the threat model, formalize flatness against it (Section 4.2), then reuse the classifier itself, without architectural change, as the enforcement mechanism during sweetword construction (Section 4.6). We call this reuse a *defensive oracle* to underline that the same object is used offensively in PassFilter and defensively here; only the surrounding pipeline changes.

Two aspects of the PassFilter setup differ from ours in ways that matter. The paper releases

model weights trained on a specific corpus and uses them to attack sweetwords produced by generators trained on that same corpus. We do not reuse those weights; we train our own CNN on a disjoint attacker corpus (Section 4.3), which raises the baseline attacker success rate but removes any shared-parameter bias between attacker and generator. The paper also evaluates flatness at the level of individual sets, whereas we report the full ASR curve $\text{ASR}(t)$ over the guess budget t , computed over the accepted sets, for every generator under a common attacker.

3.6 The HBAAT Survey

Chakraborty et al. [Cha+23] survey the honeyword literature and propose a framework, *honeyword-based authentication techniques* (HBAAT), that organizes existing schemes along three axes: generation method, distributed security model, and evaluation metric. Their treatment of flatness is compatible with, and formalizes, the definition given by Juels and Rivest [JR13]; they extend it to ε -flatness and connect it to the two further security notions mentioned in Section 2.3.

Their attacker taxonomy runs from frequency-based rankers through PCFG and Markov models to neural language models. Neural *distinguishers* of the kind introduced by Dani, McCulloh, and Saxena [DMS24] appear implicitly in the survey’s discussion of adversarial machine learning but are not folded into the flatness definition itself. The refinement in Section 4.2 can be read as that integration.

3.7 Position of This Thesis

The construction developed here sits at the intersection of the two threads. From the generator thread it inherits the ten methods of Section 4.5 unmodified. From the distinguisher thread it inherits the CNN classifier of Dani, McCulloh, and Saxena [DMS24], again without architectural change. What is new is the pipeline combining them: the classifier is trained on a disjoint corpus and reused as the oracle deciding which sweetword sets get stored, and the flatness conditions it enforces are stated over the set-wide induced distribution of Section 4.2 rather than over individual candidates.

Three observations situate the contribution.

1. Prior generators are evaluated against attackers they themselves inspired. Frequency-based generators are evaluated against frequency rankers, representation-learning generators against Markov or PCFG rankers. This thesis holds the attacker class fixed, at the CNN of Dani, McCulloh, and Saxena [DMS24], and lets the generators

vary; that common attacker is what makes the ten methods comparable on a single ASR axis. Robustness to the choice of attacker is examined separately (Section 5.8) by re-evaluating the accepted sets against a mixed-corpus CNN and a frequency ranker.

2. Prior work on neural distinguishers stops at demonstrating that flatness is violated, without proposing a construction that restores it. This thesis proposes one and measures how far it gets.
3. Filtering ideas already exist in the honeyword literature, but they operate at the candidate level with non-learned filters such as frequency pruning and dictionary matching. The two-stage filter of Section 4.6 uses a learned filter and, in its set-level stage, constrains the distribution rather than the individual candidates. That combination is what lets it enforce ε -flatness in the sense of Equation (4.7).

The rest of the thesis develops these three points. Chapter 4 specifies the construction and Chapter 5 measures its effect on ASR across the ten generators.

Chapter 4

Methodology

This chapter describes the construction we use to enforce set-level flatness against a learned attacker. Section 4.1 states the pipeline and its main design choices. Section 4.2 formalizes the attacker-induced distribution and the flatness metrics computed on top of it. Section 4.3 describes the corpora and the disjoint oracle/attacker split. Section 4.4 specifies the character-level CNN that acts as defensive oracle during filtering and as adversary during evaluation. Section 4.5 describes the ten honeyword generators we benchmark. Section 4.6 presents the hybrid flatness-filtering algorithm. Section 4.7 describes the evaluation protocol and the rational-attacker floor applied to reported ASR. Section 4.8 lists the locked parameters used by every experiment in Chapter 5.

4.1 Overview

For each real password w^* , a generator produces $k - 1$ candidate honeywords which are combined with w^* into a sweetword set $S = \{w_1, \dots, w_k\}$ of size $k = 20$. Two independently trained character-level CNNs touch S at different points in the pipeline. The first, the *oracle*, runs at storage time inside the filtering algorithm, deciding whether S satisfies the flatness conditions of Section 4.2 and triggering regeneration when it does not. The second, the *attacker*, runs at evaluation time to measure the attacker success rate on the accepted sets. The two networks share an architecture but are trained on disjoint corpora with different random seeds (Section 4.4), which is what lets us report ASR against an attacker that is not the model the filter was tuned against.

Filtering happens once, at password storage time, and never during authentication. The construction is generator-agnostic: we apply the same filter with the same thresholds on top of ten distinct generators, one of which (HoneyGen Baseline) performs no filtering at

all and serves as the reference for measuring the filter’s effect.

4.2 Attacker-Relative Set-Level Flatness

4.2.1 Sweetword Set and Attacker Model

A sweetword set is a size- k multiset $S = \{w_1, \dots, w_k\}$ in which exactly one element w^* is the real password and the remaining $k - 1$ are honeywords from a generator. The attacker A is a binary classifier trained on labelled examples (w, y) with $y = 1$ for real passwords and $y = 0$ for honeywords. For every candidate w_i it outputs an independent score

$$s_i = s(w_i) = P_A(y = 1 \mid w_i) \in [0, 1]. \quad (4.1)$$

These are per-candidate probabilities from a binary classifier, not a distribution over S . In general $\sum_i s_i \neq 1$, and the classifier sees each w_i without any other member of S .

4.2.2 Set-Level Induced Distribution

The attacker’s real decision problem is comparative: given all of S , pick one candidate. We model that comparison by renormalizing the per-candidate scores into a distribution over S :

$$q_i = \frac{s_i}{\sum_{j=1}^k s_j}, \quad \sum_{i=1}^k q_i = 1. \quad (4.2)$$

The vector $q = (q_1, \dots, q_k)$ is the *attacker-induced distribution* on S . Under ideal flatness q is uniform, so $q_i = 1/k$ for every i . We write q_{real} for the entry corresponding to w^* and use it as the central quantity the filter constrains.

4.2.3 Flatness Metrics

Four quantities computed from the pair (s, q) characterize how far S deviates from flatness under A .

Rank-based flatness. Order the candidates by s_i in decreasing order and let $\text{rank}(w^*)$ be the position of w^* . We define

$$F_{\text{rank}} = \frac{\text{rank}(w^*) - 1}{k - 1} \in [0, 1]. \quad (4.3)$$

A value of 0 means the attacker ranks w^* first, the worst case for the defender; a value of 1 means it ranks w^* last. Under a uniform ranker the expected value is $1/2$.

Normalized entropy. The Shannon entropy of q divided by its maximum $\log k$:

$$F_{\text{ent}} = \frac{H(q)}{\log k} = -\frac{1}{\log k} \sum_{i=1}^k q_i \log q_i. \quad (4.4)$$

$F_{\text{ent}} = 1$ exactly when q is uniform. Entropy captures global smoothness of q but says nothing about which entry is largest, and a set can have F_{ent} close to 1 while still giving w^* the top rank. We therefore treat F_{ent} as a secondary stabilizing condition rather than a primary one.

Gap metrics. Let $s_{(1)} \geq s_{(2)} \geq \dots \geq s_{(k)}$ be the sorted raw scores. The top gap

$$\Delta_{\text{top}} = s_{(1)} - s_{(2)} \quad (4.5)$$

measures how strongly the attacker prefers its top candidate over the runner-up, and the real gap

$$\Delta_{\text{real}} = s_{(1)} - s(w^*) \quad (4.6)$$

measures how far the real password falls below the top score. Both are computed on raw scores rather than on q , since ranking is invariant under the positive rescaling of Equation (4.2) and the raw scores keep the absolute magnitude of the classifier’s preference.

ε -flatness. Let $q_{\max} = \max_i q_i$ be the largest entry of the induced distribution. The set S is ε -flat under A if

$$q_{\max} \leq \frac{1}{k} + \varepsilon. \quad (4.7)$$

Setting $\varepsilon = 0$ forces $q_{\max} \leq 1/k$, which together with $\sum_i q_i = 1$ recovers exact uniformity of the whole distribution; larger ε relaxes the bound. Since $q_{\text{real}} \leq q_{\max}$, the constraint upper-bounds the attacker’s posterior on every sweetword including the true credential, and it is the primary security constraint the filter enforces. This additive slack on the maximum entry differs from the classical multiplicative bound $(1 + \varepsilon)/k$ of Section 2.3. The two coincide at $\varepsilon = 0$, but for $\varepsilon > 0$ the additive form puts the threshold at $1/k + \varepsilon$ rather than $(1 + \varepsilon)/k$, so the same numerical ε is a looser bound here. We use the additive form because it acts directly on the peak of the attacker-induced distribution over the set, whereas the classical bound is stated on the attacker’s first-guess success probability.

Attacker success rate. Given the ranking induced by s , the attacker success rate at rank t is

$$\text{ASR}@t = \Pr[\text{rank}(w^*) \leq t], \quad (4.8)$$

with the probability taken over the sample of evaluation sets. Under ideal flatness $\text{ASR}@t = t/k$. We report $\text{ASR}@1$, $\text{ASR}@3$, and $\text{ASR}@5$ throughout Chapter 5, with $\text{ASR}@5$ as the primary security metric.

4.2.4 Operational Definition of Flatness

The set S is declared flat under A if it satisfies Equation (4.7) together with two auxiliary conditions on F_{ent} and Δ_{top} . The three thresholds and their justification appear in Section 4.8. Sets failing any of the three are rejected by the filter of Section 4.6.

4.3 Corpora and Splits

4.3.1 Sources

The corpus is the cleaned RockYou leak, from which we draw disjoint oracle and attacker subsets after deduplication, encoding validation, and truncation to the CNN’s input length. The oracle corpora are the three training sizes $N_{\text{train}} \in \{2000, 5000, 10000\}$ used throughout Chapter 5; the attacker corpus is a fixed disjoint subset of 5000 passwords, split 80/10/10 and reused unchanged for every oracle size (the training composition is detailed in Section 4.4). All entries are unprocessed leaked passwords, with no case folding, character substitution, or semantic normalization beyond the length truncation. Extending the benchmark to a second corpus such as *darkc0de* is left as future work (Chapter 6).

4.3.2 Disjoint Oracle and Attacker Corpora

The pipeline uses two disjoint corpora. The *oracle corpus* supplies the real passwords w^* from which sweetword sets are built and provides the training data for the oracle CNN used by the filter; a separate oracle is trained for each generator size N_{train} . The *attacker corpus*, drawn from the same source but disjoint from the oracle corpus, provides the training data for the attacker CNN that measures ASR on the accepted sets. Unlike the oracle, the attacker is trained once on the training split of a fixed 5000-password corpus (Section 4.4) and reused across every generator size, so one consistent adversary evaluates all of them. Because it is trained on passwords neither the generators nor the oracle have seen, its evaluation performance reflects generalization to unseen inputs rather than memorization

of the sets it will later rank. This raises the baseline ASR compared with a shared-corpus setup, but the resulting numbers are the ones that matter for a real deployment.

4.3.3 Set-Level Splits

Within the oracle corpus, real passwords are partitioned into training, validation, and test subsets. Sweetword sets are constructed for every real password in each subset, and all sets are concatenated into a single evaluation stream with globally unique set identifiers. This `EVAL_FULL = True` configuration avoids staircase artifacts in the ASR curve at small evaluation sizes, which would otherwise arise from evaluating only the tail of a single split.

4.4 The CNN Discriminator

4.4.1 Architecture

The classifier used for both roles is a character-level convolutional network. Each password is truncated or right-padded to a fixed length and mapped through a character embedding of dimension 64. Two one-dimensional convolutional layers with ReLU activations extract local n -gram features, followed by max-pooling to reduce dimensionality. A fully connected layer projects to a two-dimensional output over which a softmax produces $P(\text{real} \mid w)$ and $P(\text{honey} \mid w)$. The architecture follows the learned-attacker paradigm of Dani, McCulloh, and Saxena [DMS24] but is trained independently and reuses no released weights.

4.4.2 Training

The classifier is trained with cross-entropy loss

$$\mathcal{L} = -(y \log s + (1 - y) \log(1 - s)). \quad (4.9)$$

Positive examples are real passwords; negative examples are honeywords from the HoneyGen Baseline generator of Section 4.5.2, which combines the FastText password model with rule-based tweaking. Training against HoneyGen negatives calibrates the classifier to the dominant family of decoys studied here and produces a strong, realistic adversary. The oracle is trained on the training split of its own N_{train} corpus. The attacker is trained once on a fixed 5000-password subset of RockYou, disjoint from every oracle corpus and split 80/10/10 into training, validation, and test partitions; its training set is built by running HoneyGen Baseline with $k = 20$ on the 4000 real passwords of the training partition, producing 19 honeywords per real password. The attacker CNN is therefore trained on

4000 real and 76,000 honeyword examples, and this single trained attacker evaluates every generator at every training size. Training uses batch size $BS_CNN = 256$ and runs for ten epochs in both roles.

4.4.3 Dual Role and Seed Separation

The two networks share the architecture and training procedure above but differ in three respects. They are trained on disjoint corpora (Section 4.3). The attacker uses a seed offset of 7777 relative to the oracle, so the two are not identical even where their training procedures coincide. And only the oracle interacts with the filter at storage time; the attacker runs only at evaluation time and never sees the acceptance decisions the oracle made.

4.4.4 Robustness Attackers

The CNN above is the attacker the filter is tuned against. To test whether the flatness it enforces survives a change of attacker, we evaluate the accepted sets against two further attackers the filter never saw. Both run only at evaluation time, on sets the filter has already accepted; neither takes part in filtering.

Frequency ranker. The frequency attacker is a classical, non-learned baseline. It builds a frequency table from the attacker corpus by counting how often each password occurs. To rank a sweetword set it scores each of the k candidates by that count, so a common password scores high and a rare or never-seen one scores low, with unseen candidates receiving the lowest score. It then guesses the candidates from most to least frequent, on the assumption that a real password is more likely to be a popular one. It uses no neural network and no training beyond counting, which is what makes it a useful control: this is the ranker-style attacker most earlier honeyword evaluations assumed, so comparing it against the CNN shows how much a learned discriminator adds over simple popularity.

Mixed-corpus CNN. The mixed-corpus attacker has the same architecture and training procedure as the oracle CNN, but its negative examples come from several generator families together, Chaffing-by-Tweaking, the Password-Model, HoneyGen, and the GAN, instead of from HoneyGen Baseline alone. Training on a broader mix stops the attacker over-fitting to one decoy family and lets it flag sets, such as the GAN’s, that a single-family CNN misses (Section 5.7).

4.5 Honeyword Generation Methods

We benchmark ten generators. four are unfiltered baselines that expose how each generation strategy behaves against the learned attacker; one GAN as a self trained generator and five are filtered variants combining a generator with the pre-screening or rejection sampling stages of Section 4.6.

4.5.1 Chaffing Family

The three *chaffing* methods derive candidates directly from w^* and perform no oracle-gated selection.

Chaffing-by-Tweaking. Rule-based edits applied to w^* : character-class substitutions on letters, digits, and symbols; suffix and prefix modifications; and short case flips. Candidates preserve the length structure of the real password.

Chaffing-with-Password-Model. Nearest neighbours of w^* in the FastText embedding space, using the skipgram model with `minCount` = 1 and `minn` = 2 trained on the oracle corpus. Candidates share sub-word structure with w^* but need not preserve its length.

Chaffing-with-Hybrid-Model. This generator combines the two streams above. For a real password w^* it draws candidates from both sources, the length-preserving rule-based edits of Chaffing-by-Tweaking and the FastText nearest neighbours of Chaffing-with-Password-Model, and forms the $k - 1$ decoys of each sweetword set from the combined pool. A set can therefore contain both surface-level variants of w^* , from the tweaking stream which keeps its length and character structure, and sub-word-similar neighbours from the password-model stream, which share its morphology but not necessarily its length. Mixing the two makes accepted sets harder to characterize with a single heuristic, since no one edit rule or embedding-distance cutoff explains all the decoys. Because both streams reuse the same underlying components as HoneyGen Baseline, the random seed governing how they are combined is offset by 5000 from the base seed, so the hybrid method produces a distinct sample rather than reproducing HoneyGen Baseline’s sets.

4.5.2 HoneyGen Family

The five *HoneyGen* methods build on the algorithm of Dionysiou, Vassiliades, and Athanassopoulos [DVA21], which combines the FastText password model with rule-based tweaking

and applies frequency-based pruning to the resulting candidate pool.

HoneyGen Baseline. The HoneyGen algorithm as described by Dionysiou, Vassiliades, and Athanasopoulos [DVA21], with no oracle interaction. This is the reference point against which every filter’s effect is measured.

HoneyGen Set-Filter. HoneyGen Baseline followed by set-level rejection sampling only. Candidate honeywords are used as produced, and the oracle decides acceptance at the sweetword-set level.

HoneyGen Prescreen-Add and Prescreen-Ratio. HoneyGen Baseline restricted to its password-model component (the `honeygen-model-only` configuration), followed by both candidate-level pre-screening and set-level rejection sampling. The Add variant uses the additive band condition $|s(w) - s(w^*)| \leq \tau_{\text{band}}$, and the Ratio variant uses the two-sided multiplicative band condition $\frac{1}{\tau_{\text{band-ratio}}} s(w^*) \leq s(w) \leq \tau_{\text{band-ratio}} s(w^*)$ with $\tau_{\text{band-ratio}} > 1$, both defined in Section 4.6.

HoneyGen Hybrid-Add and Hybrid-Ratio. Identical to the Prescreen variants except that the candidate stream is the full HoneyGen hybrid, password model plus tweaking, rather than password-model only. Both use the two-stage filter.

4.5.3 GAN

A generative adversarial network trained separately for each training-set size. Candidates are drawn from the trained generator, then oversampled by a factor of roughly five, deduplicated, and stripped of malformed entries; a random subset of the survivors becomes the $k - 1$ honeywords. The GAN is not gated by the oracle. Its results in Chapter 5 come with a qualification we discuss there.

4.6 Hybrid Flatness-Filtering Algorithm

The filter has two stages. Candidate-level pre-screening removes individually distinguishable decoys before a sweetword set is assembled, and set-level rejection sampling discards sets whose induced distribution violates one of the flatness conditions from Section 4.2. Both stages use the oracle CNN of Section 4.4.

4.6.1 Candidate-Level Pre-Screening

For each candidate honeyword w , the oracle produces a score $s(w) \in [0, 1]$. The pre-screening condition compares this score to the score $s(w^*)$ of the real password within a band around it. The *additive* band accepts w when

$$|s(w) - s(w^*)| \leq \tau_{\text{band}}, \quad (4.10)$$

and the *multiplicative* band accepts w when its score lies within a factor $\tau_{\text{band-ratio}}$ of $s(w^*)$ on either side,

$$\frac{1}{\tau_{\text{band-ratio}}} s(w^*) \leq s(w) \leq \tau_{\text{band-ratio}} s(w^*), \quad \tau_{\text{band-ratio}} > 1. \quad (4.11)$$

The two variants correspond to the Prescreen and Hybrid families of Section 4.5. Both bands are two-sided: they remove candidates the oracle is confident about in either direction, whether obviously honeyword-looking with a score far below $s(w^*)$, or obviously real-looking with a score far above it, since either kind stands out in the induced distribution.

The scale of s is worth stating explicitly. The oracle is a binary classifier, so $s(w)$ is its confidence that w is a real password, not a probability over the password space; a value of 0.2 means the classifier is 20% confident, which is an ordinary output rather than an implausibly popular password. Because the attacker corpus contains one real password per $k = 20$ candidates, the classifier’s base rate is $1/k = 0.05$, and measured scores for real passwords cluster around 0.20–0.25 with a fifth percentile near 0.12. One practical adjustment is still needed for the multiplicative band: an undertrained oracle occasionally scores a real password close to zero, which would shrink the band to almost nothing and reject every candidate. The band is therefore built around $\max(s(w^*), 0.05)$, with the floor set at the base rate. In our runs that floor binds on fewer than 1% of passwords, so for almost every set the band is centred on $s(w^*)$ itself.

4.6.2 Set-Level Rejection Sampling

After $k - 1$ candidates survive pre-screening, the sweetword set $S = \{w^*\} \cup \{w_1, \dots, w_{k-1}\}$ goes to the oracle, which produces the score vector $s = (s_1, \dots, s_k)$ and, via Equation (4.2), the induced distribution q . The set is accepted if all three of the following hold:

$$q_{\text{max}} \leq \frac{1}{k} + \varepsilon_{\text{max}}, \quad (4.12)$$

$$F_{\text{ent}} \geq 1 - \delta_H, \quad (4.13)$$

$$\Delta_{\text{top}} \leq \text{gap}_{\text{max}}. \quad (4.14)$$

If any condition fails, the whole set is discarded and a fresh one generated. Regeneration continues until acceptance or until the regeneration cap $R_{\max}$ is reached, at which point the best set observed so far, measured by minimum $q_{\max}$, is retained.

One design limitation of this test is worth stating. The predicate caps only the largest entry $q_{\max}$ and imposes no lower bound on any entry, so a set in which some honeyword receives an anomalously low q can still pass even though the distribution is not perfectly uniform. In practice the asymmetry does not affect ASR, since ASR responds to the top of the ranking rather than the tail; we return to this in Chapter 5.

4.6.3 Algorithm

The combined procedure appears in Algorithm 1. The pseudocode uses a generic band operator $\text{band}(\cdot)$ standing for either the additive or the ratio variant of the pre-screening rule.

Algorithm 1 Hybrid Flatness Filtering

Require: real password w^* , generator G , oracle A , thresholds τ_{band} , $\varepsilon_{\max}$, δ_H , $\text{gap}_{\max}$, cap $R_{\max}$

Ensure: sweetword set S

```

1:  $r \leftarrow 0$ ;  $S^{\text{best}} \leftarrow \emptyset$ ;  $q^{\text{best}} \leftarrow +\infty$ 
2: while  $r < R_{\max}$  do
3:    $C \leftarrow G.\text{GENERATE}(w^*, N_{\text{over}})$  ▷ oversampled candidate pool
4:    $C \leftarrow \{w \in C : \text{BAND}(s(w), s(w^*))\}$  ▷ candidate pre-screening
5:   if  $|C| < k - 1$  then
6:      $r \leftarrow r + 1$ ; continue
7:   end if
8:    $S \leftarrow \{w^*\} \cup \text{TAKEFIRST}(C, k - 1)$ 
9:   compute  $s_i = A(w_i)$  and  $q_i$  via Equation (4.2)
10:  compute  $q_{\max}$ ,  $F_{\text{ent}}$ , and  $\Delta_{\text{top}}$ 
11:  if  $q_{\max} \leq 1/k + \varepsilon_{\max}$  and  $F_{\text{ent}} \geq 1 - \delta_H$  and  $\Delta_{\text{top}} \leq \text{gap}_{\max}$  then
12:    return  $S$ 
13:  end if
14:  if  $q_{\max} < q^{\text{best}}$  then
15:     $S^{\text{best}} \leftarrow S$ ;  $q^{\text{best}} \leftarrow q_{\max}$ 
16:  end if
17:   $r \leftarrow r + 1$ 
18: end while
19: return  $S^{\text{best}}$ 

```

4.6.4 Complexity

Let C_{CNN} be the cost of one forward pass through the oracle and $\mathbb{E}[R]$ the expected number of regeneration attempts per password. The expected cost of producing one accepted sweetword set is

$$\mathcal{O}(\mathbb{E}[R] \cdot k \cdot C_{\text{CNN}}), \quad (4.15)$$

plus the generator’s own cost. Since C_{CNN} dominates the arithmetic cost of any generator used here, the filter’s total cost is close to $\mathbb{E}[R] \cdot k$ oracle forward passes per password. Batched scoring keeps this modest for $k = 20$ and the R_{max} values of Section 4.8, and filtering happens once at storage time rather than at every authentication.

4.7 Evaluation Protocol

4.7.1 Design

Each generator is trained once at a fixed training size N_{train} , and every evaluation set comes from that single trained generator. ASR@ t is then computed over a test set of sweetword sets, restricted to the sets the filter accepted (Section 4.6). Because the generator is fixed for a given N_{train} , the reported figures compare methods on the same footing rather than confusing a change of generator with a change of measurement. We report results at three training sizes, $N_{\text{train}} \in \{2000, 5000, 10000\}$, each evaluated on a test set of 2000 sweetword sets.

4.7.2 Rational Attacker Floor

The CNN attacker’s raw ASR@ t , the success of following the CNN’s t highest-scored candidates, can fall *below* the random-guessing baseline t/k on some sets. That is not a sign the set is safe. It means the CNN’s ranking is *anti-informative*: it places the real password near the *bottom* of the order, so following its top guesses does worse than chance. A rational attacker who noticed this would stop following the CNN and invert the strategy, guessing the candidates the CNN scores *lowest*, which then succeeds far more often than chance.

We therefore report the success of a rational attacker that uses the CNN ranking in whichever direction helps, falling back on random guessing when neither direction is informative:

$$\text{ASR}_{\text{reported}}(t) = \max(\text{ASR}_{\text{CNN}}(t), \text{ASR}_{\overline{\text{CNN}}}(t), t/k), \quad (4.16)$$

where $\text{ASR}_{\text{CNN}}(t)$ counts the sets whose real password is among the CNN’s top t candidates, $\text{ASR}_{\overline{\text{CNN}}}(t)$ counts those whose real password is among the CNN’s *bottom* t candidates, and t/k is plain random guessing. A genuinely flat generator drives all three terms to t/k , so the reported value is t/k . A generator whose real password the CNN systematically ranks low, notably the GAN of Section 5.7, is caught by the inverted term, which reports the high success such a set really allows instead of masking it at t/k . This is the value reported throughout Chapter 5; the GAN-focused comparisons in Section 5.7 additionally break it into its follow-the-CNN and inverted components and report the raw per-attacker success, so the inversion behind the GAN’s number is visible rather than hidden.

4.8 Experimental Protocol and Parameters

4.8.1 Locked Parameters

The following parameters are fixed for every experiment in Chapter 5. The flatness thresholds were set in two stages. An initial, strict configuration ($\varepsilon_{\max} = 0.02$, $\delta_H = 0.04$, $\text{gap}_{\max} = 0.01$) came from the theoretical rationale in Section 4.2 and preliminary runs on the training split. As reported in Section 4.8.2, that configuration drove the ratio-based filters to acceptance rates below 10%, at which point the filtered methods can no longer be compared meaningfully against the unfiltered baselines. We therefore adopted the relaxed configuration listed below, which restores usable acceptance across all filters at every corpus size. None of the thresholds was tuned on the evaluation split; the revision came entirely from the acceptance-rate diagnostic on the training split.

Symbol	Value	Role
k	20	Sweetword set size
$\varepsilon_{\max}$	0.05	ε -flatness bound (see note below)
δ_H	0.10	Entropy slack, $F_{\text{ent}} \geq 1 - \delta_H$
$\text{gap}_{\max}$	0.05	Top-gap threshold
τ_{band}	0.15	Additive pre-screening band
$\tau_{\text{band-ratio}}$	2.0	Multiplicative pre-screening band
$R_{\max}$	50	Regeneration cap
BS_CNN	256	CNN training batch size
Epochs (oracle)	10	CNN oracle training epochs
Epochs (attacker)	10	CNN attacker training epochs
SEED	42	Base random seed
Attacker seed offset	7777	Added to base seed for attacker
Hybrid model seed offset	5000	Added to base seed for Hybrid-Model

A single value of $\varepsilon_{\max} = 0.05$ is used throughout, both in the definition of ε -flatness (Equation (4.7)) and in the filter’s acceptance predicate (Section 4.6), which share the additive form $q_{\max} \leq 1/k + \varepsilon_{\max}$.

4.8.2 Threshold Selection and the Accept-Rate / ASR Trade-off

The flatness thresholds control a trade-off between two quantities a deployed countermeasure has to satisfy at the same time. A stricter gate produces flatter accepted sets and lowers ASR, but rejects a larger share of passwords; each rejected password hits the regeneration cap without an accepted set and is excluded from the reported ASR, which is therefore conditional on acceptance. The accept rate measures how often the filter succeeds; the ASR measures how flat the sets are when it does.

Under the initial thresholds the ratio-based filters accepted fewer than 10% of candidate passwords at $N_{\text{train}} \in \{5000, 10000\}$, with Set-Filter reaching 1.2% at $N_{\text{train}} = 5000$. At that level their ASR describes a small, self-selected minority of passwords rather than the method as a deployed defence. The adopted thresholds raise every filter above 25% acceptance at every corpus size (Chapter 5), at the cost of a modest increase in ASR on the accepted sets. The choice is deliberate: a filter protecting a single-digit percentage of users is not a usable countermeasure however flat its accepted sets are, so acceptance is treated as a first-class deployability requirement rather than a secondary diagnostic.

Methods are therefore compared in two stages in Chapter 5. A method must first clear a usable acceptance threshold; only among those that do is the flattest, lowest-ASR method preferred. Under this criterion the ratio-based filters are the recommended configurations, delivering the flattest sets among the methods that stay deployable, whereas the additive filters accept more passwords but sit closer to the unfiltered baseline in flatness.

4.8.3 Metrics and Statistical Treatment

For each generator and each N we report ASR@1, ASR@3, and ASR@5 using Equation (4.16), with ASR@5 as the primary metric. Alongside each ASR value we report the mean and median of F_{rank} , F_{ent} , q_{real} , and Δ_{top} over the evaluation sets, together with the number of accepted sets for the filtered generators. Confidence intervals at small N are wide, roughly ± 20 percentage points at $N \leq 500$, and we draw no conclusions from small- N comparisons falling inside that margin.

4.8.4 Robustness Check

The primary corpus for all experiments is RockYou and the primary attacker is the HoneyGen-tuned CNN. Robustness is assessed by re-evaluating the accepted sets against two further attacker classes, a mixed-corpus CNN trained on several decoy families and a frequency ranker, as reported in Section 5.8. Extending the benchmark to a second corpus such as *darkcode* is left as future work (Chapter 6).

Chapter 5

Results

This chapter reports the experiments defined in Section 4.7. Section 5.1 recaps the setup and how to read the tables. Section 5.2 shows how vulnerable the unfiltered honeyword generators are to the CNN attacker of Section 4.4. Section 5.3 isolates the effect of the two-stage filter of Section 4.6 on ASR@ t and on the flatness criteria of Section 4.2. Section 5.4 reports the acceptance rates and regeneration counts and develops the deployability-first reading. Section 5.5 confirms that the ordering of methods holds across training-set sizes. Section 5.7 isolates a distributional artifact of the GAN. Section 5.8 evaluates the accepted sets against a frequency attacker as a cross-attacker check. Section 5.9 draws the picture together.

5.1 Setup

All experiments run on the cleaned RockYou corpus. For each training size $N_{\text{train}} \in \{2000, 5000, 10000\}$ a separate oracle and generator are trained, and every trained generator produces sweetword sets evaluated on a test set of 2000 sets (Section 4.7). The attacker CNN is trained once on a disjoint 5000-password attacker corpus with HoneyGen-Baseline negatives and reused unchanged across all training sizes and generators (Section 4.4), so one consistent adversary evaluates every method. Parameter values are those of Table 5.1 in Section 4.8, that is, the relaxed thresholds adopted in Section 4.8.2.

The primary security metric is ASR@5: the fraction of sets on which the attacker’s five highest-ranked guesses include the real password. Under ideal flatness $\text{ASR}@5 = 5/k = 0.25$ for $k = 20$, and higher values say how much better than random guessing the attacker does. We also report the flatness criteria: F_{rank} , the normalised rank of the real password, where 0.5 is the ideal middle; F_{ent} , the normalised entropy of the induced distribution,

Symbol	Meaning	Value
k	Sweetword set size	20
$\varepsilon_{\max}$	ε -flatness bound ($q_{\max} \leq 1/k + \varepsilon_{\max}$)	0.05
δ_H	Entropy slack ($F_{\text{ent}} \geq 1 - \delta_H$)	0.10
$\text{gap}_{\max}$	Top-gap threshold	0.05
τ_{band}	Additive pre-screening band	0.15
$\tau_{\text{band-ratio}}$	Multiplicative pre-screening band	2.0
$R_{\max}$	Regeneration cap	50
BS_CNN	CNN training batch size	256
Epochs (oracle / attacker)	CNN training epochs	10 / 10
Epochs (GAN final / quick)	GAN training epochs	60 / 20
Epochs (FastText final / quick)	FastText training epochs	100 / 20

Table 5.1: Adopted parameter values for all experiments in Chapter 5. The flatness thresholds are the relaxed configuration of Section 4.8.2.

where close to 1 is uniform; and F_ε , the ε -flatness pass rate. Table 5.1 lists every fixed parameter. Unless stated otherwise, the detailed single-training-size tables use $N_{\text{train}} = 10000$, the largest and most stable corpus; Section 5.5 shows the smaller sizes reproduce the same ordering.

5.2 Unfiltered Baselines Under the CNN Attacker

The four unfiltered strategies (Chaffing-by-Tweaking, Chaffing-with-Password-Model, Chaffing-with-Hybrid-Model, HoneyGen Baseline) are all highly distinguishable to the CNN attacker. At $N_{\text{train}} = 10000$ they report ASR@5 of 0.678, 0.755, 0.743, and 0.746 (Table 5.2), roughly three times the random-guessing rate of 0.25. Two points follow. The four strategies cluster tightly: the three model-based generators sit within 0.02 of one another, and the simplest rule-based generator, Chaffing-by-Tweaking, is only about 0.07 lower. And the more elaborate statistical machinery of the hybrid and HoneyGen generators does *not* translate into flatness against a learned discriminator. Their F_{rank} values sit near 0.169, so the real password lands near the top of the twenty candidates on the average set. This is the pattern that motivates the filter.

Method	$N_{\text{train}}=2000$			$N_{\text{train}}=5000$			$N_{\text{train}}=10000$		
	@1	@3	@5	@1	@3	@5	@1	@3	@5
Chaffing-by-Tweaking	0.290	0.544	0.679	0.287	0.539	0.678	0.281	0.540	0.678
Chaffing-with-Password-Model	0.303	0.615	0.759	0.300	0.617	0.767	0.304	0.600	0.755
Chaffing-with-Hybrid-Model	0.341	0.611	0.756	0.338	0.620	0.752	0.337	0.603	0.743
HoneyGen Baseline	0.339	0.614	0.752	0.336	0.619	0.761	0.346	0.604	0.746
HoneyGen + Set-Filter	0.305	0.583	0.728	0.115	0.323	0.518	0.179	0.438	0.610
HoneyGen + Prescreen-Add	0.164	0.412	0.588	0.109	0.299	0.470	0.144	0.368	0.525
HoneyGen + Prescreen-Ratio	0.083	0.262	0.417	0.071	0.205	0.341	0.071	0.217	0.341
HoneyGen + Hybrid-Add	0.222	0.473	0.641	0.093	0.299	0.462	0.141	0.364	0.510
HoneyGen + Hybrid-Ratio	0.139	0.350	0.503	0.083	0.172	0.285	0.050	0.167	0.293
GAN (Self-Trained)	0.101	0.222	0.331	0.565	0.718	0.774	0.745	0.832	0.872
Random guessing (t/k)	0.050	0.150	0.250	0.050	0.150	0.250	0.050	0.150	0.250

Table 5.2: ASR@ t under the CNN attacker for all ten methods at three training sizes, reported as the rational-attacker value of Equation (4.16); no entry falls below the random-guessing baseline t/k . Columns are ASR@1, ASR@3, ASR@5. Each block is evaluated on a test set of sweetword sets produced by the generator trained at that N_{train} .

5.3 Effect of Flatness Filtering

Table 5.2 reports ASR@1, ASR@3, and ASR@5 for all ten methods at the three training sizes, and Table 5.3 consolidates the $N_{\text{train}} = 10000$ column with the flatness criteria and the filtering cost. Reading the ASR@5 column at $N_{\text{train}} = 10000$, the methods separate into three families.

Unfiltered baselines. The four baselines occupy the vulnerable band, ASR@5 between 0.678 and 0.755, and are the reference against which the filters are measured.

Additive filters. Set-Filter, Prescreen-Add, and Hybrid-Add apply the set-level acceptance test of Section 4.6, either alone in the case of Set-Filter or with a symmetric additive band on the candidate scores. They cut ASR@5 to between 0.510 and 0.610, a substantial reduction that still leaves the attacker well above the random-guessing rate: on roughly half the accepted sets the CNN’s top five guesses contain the real password. Their F_{rank} rises to about 0.296, part-way toward the ideal 0.5.

Method	ASR@5	F_{rank}	F_{ent}	F_{ε}	Accept	Tries
Chaffing-by-Tweaking	0.678	0.208	0.984	0.207	100%	1.00
Chaffing-with-Password-Model	0.755	0.171	0.971	0.125	100%	1.00
Chaffing-with-Hybrid-Model	0.743	0.171	0.970	0.132	100%	1.00
HoneyGen Baseline	0.746	0.169	0.970	0.130	100%	1.00
HoneyGen + Set-Filter	0.610	0.235	0.979	0.189	54.5%	3.51
HoneyGen + Prescreen-Add	0.525	0.296	0.990	0.329	84.6%	1.00
HoneyGen + Prescreen-Ratio	0.341	0.422	0.992	0.505	54.1%	1.00
HoneyGen + Hybrid-Add	0.510	0.296	0.989	0.328	83.3%	1.26
HoneyGen + Hybrid-Ratio	0.293	0.462	0.991	0.569	47.9%	2.68
GAN (Self-Trained)	0.872	0.920	0.989	0.958	100%	1.00

Table 5.3: Consolidated view at $N_{\text{train}} = 10000$ under the CNN attacker: security (ASR@5, lower is flatter), flatness criteria (F_{rank} and F_{ε} near 0.5 and high F_{ent} indicate near-uniform sets), and cost (accept rate and average regeneration attempts). Unfiltered methods have no filter and are shown at 100% / 1.00.

Ratio filters. Prescreen-Ratio and Hybrid-Ratio use the two-sided multiplicative band $\frac{1}{\tau_{\text{band-ratio}}} s(w^*) \leq s(w) \leq \tau_{\text{band-ratio}} s(w^*)$ during candidate screening. They are the strongest genuine defence, pushing ASR@5 down to 0.341 (Prescreen-Ratio) and 0.293 (Hybrid-Ratio) and lifting F_{rank} to 0.422 and 0.462, close to the ideal middle value of 0.5 that characterises a near-uniform ranking. Unlike the GAN (Section 5.7), the ratio filters reach low ASR *and* a central F_{rank} at once, which is what separates genuine flatness from inverted distinguishability.

Table 5.3 makes the ordering explicit and adds each filter’s cost. The relative ordering of the six filtered methods is stable across ASR@1, ASR@3, and ASR@5: ratio variants beat their additive counterparts, and Set-Filter, which uses only set-level rejection without candidate-level pre-screening, is the weakest filtered method. That confirms the pre-screening stage contributes rejection capability the set-level test alone does not recover. The table also lists the GAN for completeness; its high ASR@5 together with an F_{rank} near 1 is the inverted-distinguishability artifact analysed separately in Section 5.7, and it should not be read as a filtered defence.

5.4 Cost of Filtering and the Deployability Gate

The security ranking in Table 5.3 is only half the picture. A filter is useful only if it accepts a workable fraction of passwords. Table 5.4 reports the acceptance rate and the average number of regeneration attempts per accepted set, for the five filtered methods at each

Method	$N_{\text{train}}=2000$		$N_{\text{train}}=5000$		$N_{\text{train}}=10000$	
	Accept	Tries	Accept	Tries	Accept	Tries
HoneyGen + Set-Filter	90.6%	1.56	25.9%	4.62	54.5%	3.51
HoneyGen + Prescreen-Add	94.8%	1.00	68.4%	1.00	84.6%	1.00
HoneyGen + Prescreen-Ratio	65.5%	1.00	38.7%	1.00	54.1%	1.00
HoneyGen + Hybrid-Add	96.5%	1.03	65.5%	1.49	83.3%	1.26
HoneyGen + Hybrid-Ratio	77.5%	1.70	29.3%	3.55	47.9%	2.68

Table 5.4: Acceptance rate and average regeneration attempts per accepted set for the five filtered HoneyGen variants under the adopted thresholds. Unfiltered methods have no filter (100% acceptance, one attempt) and are omitted. Every filtered method now exceeds 25% acceptance at every training size.

training size, under the adopted thresholds.

Two readings follow. There is a clear efficiency–security trade-off: the additive filters accept the large majority of candidate sets, between 65.5% and 96.5% at $N_{\text{train}} = 2000$, in close to a single attempt, whereas the ratio filters, the strongest defence, accept a smaller fraction and cost more regeneration attempts. This is the trade-off introduced in Section 4.8.2: a stricter effective gate buys flatness at the price of acceptance.

Second, and decisive for the recommendation, the adopted thresholds keep every filtered method above the 25% acceptance mark at every training size. Under the initial strict thresholds the ratio filters fell below 10% acceptance, with Set-Filter reaching 1.2% at $N_{\text{train}} = 5000$, at which point their reported ASR described only a small, self-selected minority of passwords and could not be compared with the unfiltered methods. We therefore read the results in two stages, following Section 4.8.2. A method must first clear a usable acceptance rate to count as deployable; only among the deployable methods is the flat-test, lowest-ASR one preferred. Under this criterion the ratio filters are the recommended configurations: they deliver the flattest genuinely flat sets while remaining deployable at every corpus size, whereas the additive filters accept more passwords but sit closer to the unfiltered baseline in flatness.

Filtering happens once per password at storage time, not on every authentication (Section 4.1). Even the highest average regeneration counts in Table 5.4, for $k = 20$, amount to a sub-second cost on a small GPU and are amortised over the lifetime of the credential. A deployment that cannot afford the ratio filter’s rejection budget can fall back to an additive filter and still recover most of the security gain over the unfiltered baseline.

5.5 Stability Across Training-Set Size

Table 5.2 reports every method at all three training sizes, and the three-family structure holds throughout. The unfiltered baselines change ASR@5 by at most about 0.02 across the three corpus sizes: the CNN’s ability to separate real passwords from HoneyGen-family decoys does not depend much on how much data the generator saw. The additive and ratio filters also keep their relative order at every size, with the ratio filters the flattest genuine defence throughout. Prescreen-Ratio moves over the range 0.417–0.341 and Hybrid-Ratio over 0.503–0.293 across the three sizes. The mild upward drift of the ratio filters at intermediate sizes reflects shrinking oracle/attacker agreement as the two networks are trained on more data, not a failure of the filter; Chapter 6 returns to this boundary of model-relative flatness.

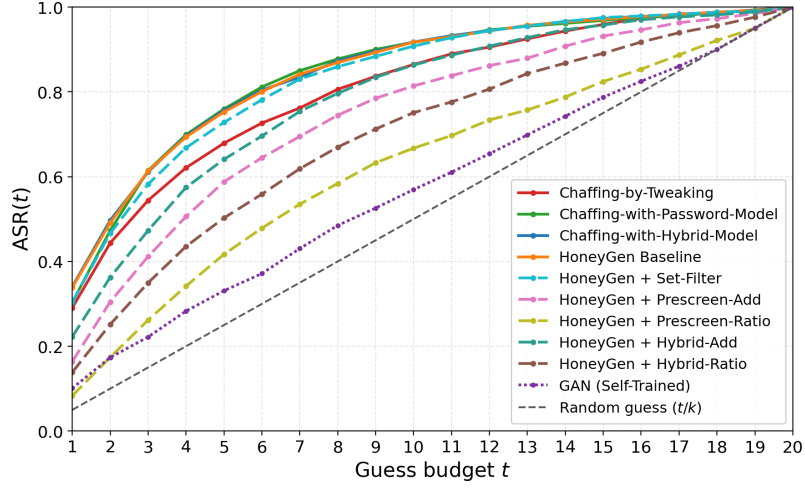
5.6 ASR Curves and Consistency Across Test-Set Size

Figure 5.1 shows the ASR curves under the CNN attacker for all ten methods at the three training sizes, each evaluated on the same test set of 2000 sweetword sets. The curves make the three-family structure visible: the unfiltered baselines sit high, the additively filtered methods fall to an intermediate band, and the ratio filters lie closest to the random-guessing diagonal. The GAN is the outlier at the top. Once the rational-attacker rule of Equation (4.16) is applied, its anti-informative ranking is exploited rather than rewarded, and it becomes the most attackable method in the benchmark (Section 5.7).

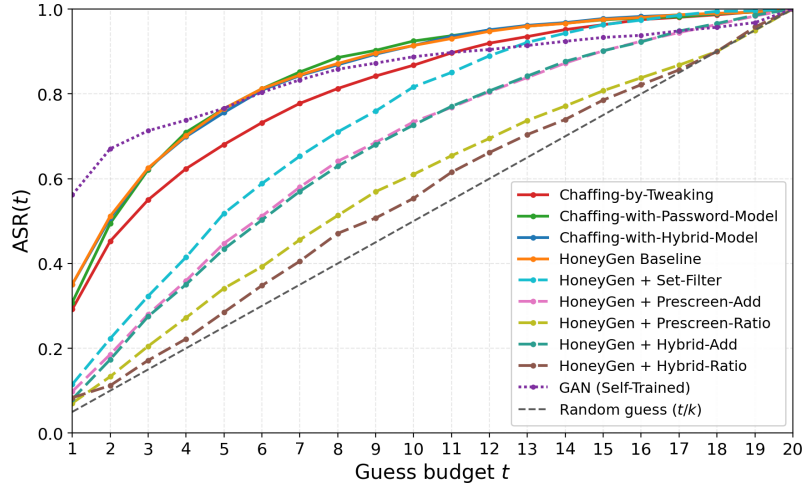
5.7 The GAN Result Requires a Caveat

The GAN’s reported $\text{ASR@5} = 0.872$ at $N_{\text{train}} = 10000$ (Table 5.2) makes it the weakest method in the benchmark, but the reason is worth spelling out, because an attacker who simply followed the CNN would conclude the opposite. Table 5.5 reports two attacker strategies side by side with F_{rank} .

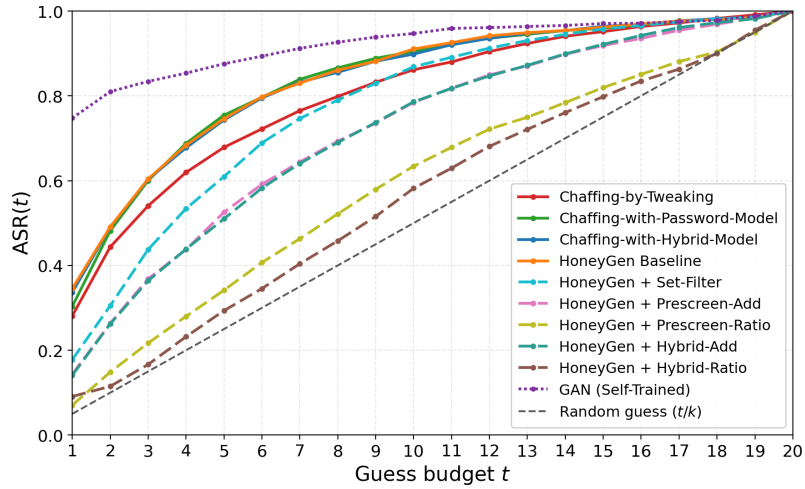
For HoneyGen Baseline, following the CNN’s top-ranked guesses succeeds 74.6% of the time and F_{rank} is 0.169: the real password sits near the top, so the attacker just follows the CNN. For the ratio filters both attacker directions land near the random-guessing rate and F_{rank} is close to the ideal 0.5; the real password is in the middle, neither direction helps, and the sets are genuinely flat. The GAN is the exception. Following the CNN succeeds only 2.5% of the time, which on its own would make the GAN look like the flattest method in the benchmark, but its F_{rank} is 0.920: the real password is ranked at or near the bottom of almost every GAN set. A rational attacker inverts and guesses the



(a) $N_{\text{train}} = 2000$.



(b) $N_{\text{train}} = 5000$.



(c) $N_{\text{train}} = 10000$.

Figure 5.1: ASR curves under the CNN attacker for all ten methods, each evaluated on the same test set of 2000 sweetword sets, at the three training sizes. Curves report the rational-attacker value of Equation (4.16), so no method falls below the random-guessing diagonal.

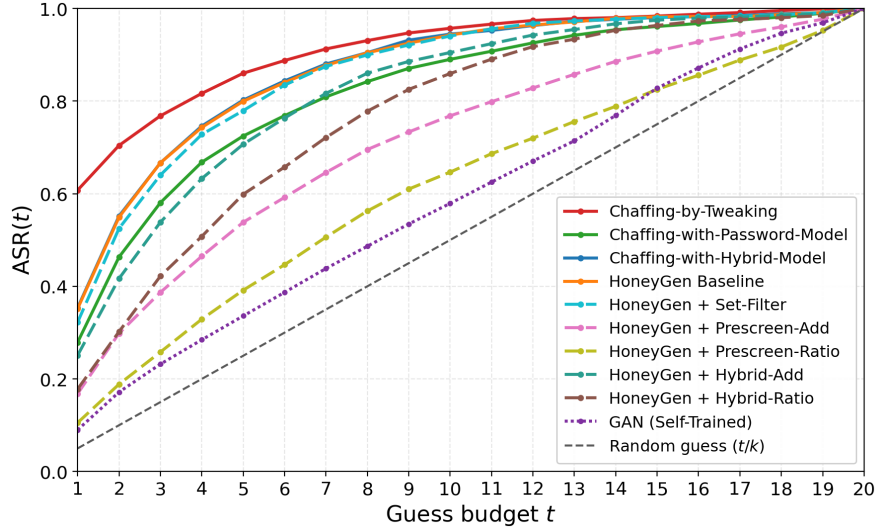


Figure 5.2: ASR curves under the mixed-corpus attacker (trained on Tweak + Password-Model + HoneyGen + GAN negatives) at $N_{\text{train}} = 2000$, test on 2000 sweetword sets.

Method	Follow-CNN ASR@5	Anti-CNN ASR@5	Reported ASR@5	F_{rank}
HoneyGen Baseline	0.746	0.038	0.746	0.169
HoneyGen + Prescreen-Ratio	0.341	0.180	0.341	0.422
HoneyGen + Hybrid-Ratio	0.293	0.202	0.293	0.462
GAN (Self-Trained)	0.025	0.872	0.872	0.920

Table 5.5: Two attacker strategies at $N_{\text{train}} = 10000$. *Follow-CNN* guesses the CNN’s top-ranked candidates, *Anti-CNN* guesses its bottom-ranked ones, and *Reported* is the rational-attacker value of Equation (4.16). For the baseline and the ratio filters, Follow-CNN is the larger of the two and F_{rank} sits near or above 0.4, so the reported value simply follows the CNN. The GAN is the exception: its F_{rank} of 0.920 puts the real password at the bottom of the set, so Follow-CNN succeeds only 2.5% of the time while Anti-CNN reaches 87%. Its low Follow-CNN score is inverted distinguishability rather than flatness.

CNN’s *lowest*-scored candidates, succeeding 87.2% of the time, which is what the two-sided floor of Equation (4.16) reports. A GAN trained on the oracle corpus samples from that corpus’s high-probability region, and the real password, itself a low-probability sample, looks like an outlier the classifier confidently rejects. This is inverted distinguishability, not flatness.

A second attacker confirms the bias without needing the inversion at all. The mixed-

Method	CNN-only ASR@5	Mixed-corpus ASR@5
Chaffing-by-Tweaking	0.679	0.860
Chaffing-with-Password-Model	0.759	0.725
Chaffing-with-Hybrid-Model	0.756	0.802
HoneyGen Baseline	0.752	0.799
HoneyGen + Set-Filter	0.728	0.779
HoneyGen + Prescreen-Add	0.588	0.538
HoneyGen + Prescreen-Ratio	0.417	0.392
HoneyGen + Hybrid-Add	0.641	0.707
HoneyGen + Hybrid-Ratio	0.503	0.599
GAN (Self-Trained)	0.212	0.336

Table 5.6: ASR@5 under the HoneyGen-only CNN attacker versus the mixed-corpus attacker (Tweak + Password-Model + HoneyGen + GAN negatives) at $N_{\text{train}} = 2000$, both following the ranking in the ordinary direction. The GAN shows the largest relative increase of any method, from 0.212 to 0.336, because the mixed attacker has seen GAN decoys during training and needs no inversion to exploit them; the filtered ratio methods shift comparatively little. The gap is far wider at $N_{\text{train}} = 5000$ (Section 5.7).

corpus attacker of Section 4.4.4 is trained on Tweak, Password-Model, HoneyGen and GAN negatives together, so GAN decoys are among the patterns it has learned to recognise. Against it the GAN is caught by following the ranking in the ordinary direction: at $N_{\text{train}} = 2000$ this reaches $\text{ASR@5} = 0.336$, against 0.212 for the HoneyGen-tuned CNN followed the same way (Table 5.6), and the gap widens with training size, reaching 0.99 at $N_{\text{train}} = 5000$ where the single-family CNN manages only 0.071. The GAN shows the largest relative increase of any method between the two attackers. Where the HoneyGen-tuned CNN has to be inverted to expose the GAN, a broader attacker simply reads it off, and either way the GAN’s apparent flatness under a naive single-family attacker is not a property of the sweetword sets it produces.

5.8 Robustness Across Attacker Class

The filter is tuned against the CNN attacker, so we check the accepted sets against the two robustness attackers of Section 4.4.4: an independent frequency ranker and a mixed-corpus CNN. Table 5.7 reports the CNN and the frequency attacker at $N_{\text{train}} = 10000$.

Three points follow. First, the frequency attacker is a substantially weaker distinguisher than the CNN: its ASR@5 on HoneyGen Baseline is 0.319 against 0.746 for the CNN. This is the gap PassFilter identified, where methods evaluated only against ranker-style

Method	CNN attacker		Freq. attacker	
	ASR@1	ASR@5	ASR@1	ASR@5
Chaffing-by-Tweaking	0.281	0.678	0.216	0.386
Chaffing-with-Password-Model	0.304	0.755	0.050	0.250
Chaffing-with-Hybrid-Model	0.337	0.743	0.083	0.319
HoneyGen Baseline	0.346	0.746	0.084	0.319
HoneyGen + Set-Filter	0.179	0.610	0.086	0.322
HoneyGen + Prescreen-Add	0.144	0.525	0.050	0.250
HoneyGen + Prescreen-Ratio	0.071	0.341	0.050	0.250
HoneyGen + Hybrid-Add	0.141	0.510	0.068	0.286
HoneyGen + Hybrid-Ratio	0.050	0.293	0.076	0.305
GAN (Self-Trained)	0.745	0.872	0.225	0.388

Table 5.7: Cross-attacker comparison at $N_{\text{train}} = 10000$, using the rational-attacker ASR of Equation (4.16). The frequency attacker separates the methods far less sharply than the CNN, and the filtered HoneyGen variants stay near its floor. The GAN is distinguishable under both: the CNN reaches 0.872 once its ranking is inverted, since the real password sits at the bottom, and the frequency attacker reaches 0.388 directly. It is flat under neither, as Section 5.7 discusses.

attackers report flatness numbers that overstate their security once a learned discriminator is admitted. Second, filtering against the CNN does not open the sets up to the frequency attacker: the filtered variants stay near the frequency floor, so there is no security trade-off between the two attacker classes. That absence of a trade-off is somewhat surprising, since tuning a filter to one attacker does not in general protect against a different one, and we have no proven explanation for it. A plausible one is that the features the CNN uses to judge how “real” a candidate looks are themselves correlated with popularity, the same common structural patterns that drive the frequency ranker, so a set the CNN cannot separate tends to be one the frequency attacker cannot separate either. We report this as an empirical observation rather than a guarantee; whether the two attacker classes genuinely align, or whether a trade-off appears against a stronger ranker, is left as future work. Third, the GAN is distinguishable to both attackers rather than flat: its frequency ASR@5 of 0.388 is the highest frequency value in the table, and under the CNN a rational attacker who inverts the ranking reaches 0.872 (Section 5.7). The GAN is caught from both directions, by the frequency ranker directly and by the CNN once its anti-informative ranking is inverted.

5.9 Discussion

The picture is consistent across training size and attacker class, with four conclusions.

First, unfiltered honeyword generators are vulnerable to a CNN attacker. The four baselines report ASR@5 around 0.68–0.76 across all three corpus sizes, about three times the random-guessing rate, and the sophistication of the generator does not translate into flatness.

Second, filtering restores flatness, most strongly in the ratio variant. The ratio filters bring ASR@5 down to 0.341–0.293 at $N_{\text{train}} = 10000$ with F_{rank} near the ideal middle, and preserve most of that reduction at smaller sizes. The additive filters achieve a moderate reduction at a much lower rejection cost.

Third, acceptance is a first-class requirement. Under the relaxed thresholds every filtered method is deployable, above 25% acceptance at every corpus size, and the recommended configurations are the ratio filters read through the two-stage criterion of Section 5.4: the flattest methods that remain deployable.

Fourth, the GAN’s apparent success is a distributional artifact. Following the CNN naively, the GAN looks like the flattest method, but its F_{rank} of 0.920 sits far from the ideal middle value of 0.5, and at the opposite extreme from the unfiltered baselines: the real password is ranked at the bottom of the set rather than the top. A rational attacker who inverts the ranking reaches ASR@5 = 0.872 (Equation (4.16)). The mixed-corpus attacker finds the same bias by following the ranking directly, and under the frequency attacker the GAN reports ASR@5 = 0.388, the highest value in that column. Any security claim treating the GAN as a strong flatness-preserving generator has to face this data. Chapter 6 takes these conclusions as its starting point.

Chapter 6

Conclusion

This thesis started from a mismatch. Honeyword-based breach detection depends on flatness, the promise that an attacker who has read the password file cannot pick the real credential out of a sweetword set with probability better than $1/k$, and every widely deployed honeyword generator has been evaluated against attackers that no longer represent the state of the art. Dani, McCulloh, and Saxena [DMS24] showed that a small character-level CNN, trained as a binary classifier on real-vs-decoy pairs, breaks flatness on essentially every scheme they tested. That left an open question: can flatness be restored against such an attacker without redesigning the generator?

The answer we develop is yes, with qualifications. The same CNN discriminator PassFilter used as an attack, reused unmodified as a defensive oracle at storage time, gates sweetword sets against a small set of measurable flatness conditions. Sets that fail are regenerated, sets that pass are stored. Neither the generator nor the attacker changes; only the pipeline combining them does. On the ten-method benchmark of Chapter 5, the two-stage filter with a multiplicative pre-screening band brings the attacker’s ASR@5 down from around 0.75 for the unfiltered baselines to 0.29–0.34 for the strongest configurations at $N_{\text{train}} = 10000$, with the real password ranked near the middle of the twenty candidates ($F_{\text{rank}} \approx 0.42$ –0.46). Under the thresholds adopted in Section 4.8.2, these ratio filters accept between 29% and 78% of passwords across corpus sizes, well inside the deployable range and far above the single-digit acceptance of the initial strict configuration.

6.1 Summary of Contributions

The contributions map to Chapter 4 and Chapter 5. First, an attacker-relative definition of flatness stated at the level of the attacker-induced distribution over a sweetword set and

characterized by four measurable quantities: F_{rank} , F_{ent} , the top gap, and an ε -flatness bound $q_{\text{max}} \leq 1/k + \varepsilon_{\text{max}}$ on the largest entry of that distribution. Second, a hybrid flatness-filtering algorithm combining candidate-level pre-screening with set-level rejection sampling, which is generator-agnostic and requires retraining neither the generator nor the attacker.

The benchmark is the third contribution: ten generation methods spanning rule-based tweaking, representation-learning HoneyGen variants, and a self-trained GAN, evaluated under a common protocol at three training sizes and against three attacker classes. The HoneyGen-tuned CNN, a mixed-corpus CNN trained on several decoy families, and a frequency ranker together separate genuine flatness from single-attacker artifacts.

6.2 What the Numbers Actually Say

Three quantitative findings are worth extracting from Chapter 5 and stating plainly for a reader in a hurry.

Unfiltered honeyword generators are highly distinguishable to the CNN. All four baselines, Tweaking, Password-Model, Hybrid-Model, and HoneyGen Baseline, report ASR@5 in the range 0.68–0.77 at every training-set size we tested, about three times the random-guessing rate of 0.25.

Flatness filtering cuts ASR@5 substantially, and the ratio band does most of the work. At $N_{\text{train}} = 10000$ the Prescreen-Ratio and Hybrid-Ratio variants push ASR@5 down to 0.341 and 0.293 with F_{rank} near the ideal middle value of 0.5, which is genuine near-uniform ranking, while the additive filters manage moderate reductions to the 0.51–0.61 range. The relative ordering of methods holds as the training corpus varies over 2000, 5000, and 10000. Because acceptance is treated as a first-class deployability requirement (Section 5.4), the recommended configurations are the ratio filters read through the two-stage criterion: the flattest methods among those that stay deployable.

The GAN’s apparent flatness is a distributional artifact and should not be read as a defence. Following the CNN’s top guesses, its ASR@5 at $N_{\text{train}} = 10000$ is essentially zero, but F_{rank} is 0.920: the real password sits near position twenty on the average set. The CNN treats the GAN’s decoys as looking *more* like real passwords than the real password itself, so a rational attacker inverts the ranking and reaches $\text{ASR@5} = 0.872$, the value the two-sided rational-attacker floor reports. That makes the GAN the weakest method in the benchmark rather than the strongest. Two independent checks confirm bias rather than flatness. Under the frequency attacker the same sets report $\text{ASR@5} = 0.388$, the highest value in that column. Under the mixed-corpus attacker, which has seen GAN decoys during training, no inversion is needed at all: following the ranking in the ordinary

direction reaches 0.99 at $N_{\text{train}} = 5000$, where the HoneyGen-tuned CNN followed the same way manages only 0.071. The artifact sharpens with corpus size; at $N_{\text{train}} = 2000$ the GAN’s F_{rank} sits closer to the middle, so the inversion grows as the generator trains on more data.

6.3 Limitations

The framework has five limitations a deployment would need to reason about explicitly.

Coverage below 100%. The most important limitation for real-world use is that the countermeasure does not protect every user. The filter stores a sweetword set only when it can certify that set as flat; when it cannot do so within the regeneration budget it gives up on that password, and that account gets no flatness-guaranteed set. At the operating point of this thesis the strongest ratio filters certify only between 29% and 78% of passwords (Table 5.4), leaving a large share of users, in the worst case the majority, without the guarantee. A deployable system has to protect 100% of accounts; it cannot simply refuse to store honeywords for the users it finds difficult. As it stands the method would need pairing with a fallback for the rejected passwords, an unfiltered set for instance, which weakens the guarantee for exactly those users. Reaching near-total coverage without sacrificing flatness is the single biggest obstacle to adoption, and we return to it as an open question below.

Model-relative guarantees. The filter enforces flatness against one specific attacker, the character-level CNN of Section 4.4. A different attacker, a larger CNN, a transformer, or a discriminator using side information beyond the password string, may recover distinguishability on sets the filter has certified. The guarantees are best read as flatness against this class of learned attacker rather than flatness in general. Section 5.8 shows they transfer partially to the frequency attacker, which is weaker; whether they transfer to stronger models is an empirical question requiring each candidate attacker to be run through the same benchmark.

One-sided set-level acceptance. The set-level acceptance test constrains only the maximum of the induced distribution q . A set in which some honeyword receives an anomalously low q would pass even though q is not close to uniform. In our data that does not translate into ASR degradation, since ASR responds to the top of the ranking rather than the tail, but the acceptance predicate could be tightened to bound both ends of q . This is a small extension of Algorithm 1 and worth doing in a production build.

Acceptance-rate cost of the ratio filter. The strongest security in the benchmark, from the Prescreen-Ratio and Hybrid-Ratio filters, comes at the lowest acceptance rates among the filtered methods, between 29% and 78% across corpus sizes against 65%–97% for the additive filters, and at a higher average number of regeneration attempts, up to about 4.6 per accepted set (Table 5.4). Under the relaxed thresholds this is no longer the crippling single-digit acceptance of the initial configuration: every filtered method clears the 25% deployability mark at every training size. The residual cost is real but falls entirely at storage time, since filtering happens once per account for the lifetime of the credential. A deployment that cannot afford the ratio filter’s rejection budget can move to an additive filter and recover most of the security gain over the unfiltered baseline at a much higher acceptance rate.

Corpus and language. All experiments use the cleaned RockYou corpus, a leak of primarily English-language, user-chosen web passwords from one consumer service. Its distributional properties, long low-probability tails, common suffixes, digit sequences at the end, are what the FastText embeddings, the GAN, and the CNN all learn to model. Whether the same numbers hold on corpora with different structure, such as enterprise credentials, non-English leaks, or machine-generated passwords, is an empirical question. Extending the benchmark to *darkc0de* as a second corpus is on the immediate next-steps list for this pipeline.

6.4 Open Questions

Beyond the immediate limitations, five research questions remain open.

The first is adversarial adaptation. As constructed, the filter is a static defence: the oracle CNN is trained once and reused. An attacker who knew the filter existed and had a sample of accepted sets could train against those sets rather than against raw generator output. Whether the generator, the oracle, and the attacker can be co-trained toward a stable equilibrium at $\text{ASR@5} \approx 0.25$ is a question this thesis does not resolve. The pieces are in place to run the experiment; the answer requires running it.

The second is transferability across a class of attackers. The current guarantees hold against one CNN. A more useful guarantee would cover a class of models, say any CNN of bounded depth and width, or any linear classifier over a fixed feature set. The filter could then use an ensemble oracle discriminating against the worst-case member of the class. Chakraborty et al. [Cha+23] discuss related notions in their survey, and connecting them to the flatness formalism of Section 4.2 is a natural extension.

The third is the theoretical relationship between the flatness metrics. F_{rank} , F_{ent} , and

ε -flatness are correlated but not equivalent, and the acceptance predicate uses all three plus the top gap. It is not obvious that this is the smallest set of conditions sufficient to bound $\text{ASR}@t$, or that all three are individually necessary. A cleaner characterization, an inequality connecting q_{real} , F_{ent} , and expected $\text{ASR}@t$, would let the filter constrain fewer quantities without giving up guarantees.

The fourth is the LLM-based generator line. The construction of Section 4.6 is generator-agnostic and would apply unchanged to sweetword sets produced by the chunk-manipulation and Metropolis–Hastings language models reviewed in Section 3.3. Whether the resulting $\text{ASR}@5$ curves match, exceed, or fall below the ratio-filtered HoneyGen numbers reported here is an experiment worth running.

The fifth is coverage. As the limitations stress, the filter protects only the passwords for which it can certify a flat set, and at the current operating point that fraction is well below 100%. A practical countermeasure has to cover every account, so the question is how to reach near-total coverage without weakening flatness. Possible routes include stronger or password-specific generators that produce flat candidates more reliably, thresholds adapted per password to the hardest cases, or a principled fallback for the residual passwords no generator can flatten. Which of these, or which combination, closes the gap is the most important question this thesis leaves open.

6.5 Closing

Honeywords are a modest defence with an outsized property: they turn a silent failure, offline password recovery from a stolen hash file, into an observable event at the authentication server. That property is worth preserving as attacker capabilities improve, and the argument here is that each new class of attacker should be reused as an enforcement tool at storage time. The classifier that would otherwise recover the real password becomes the gatekeeper that keeps any sweetword set with a recoverable real password from being stored. The reversal is small and the numbers are moderate, but the direction feels right: as long as attackers keep getting better, defenders can keep the same tools, pointed the other way.

Bibliography

- [Boj+17] Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov. “Enriching Word Vectors with Subword Information”. In: *Transactions of the Association for Computational Linguistics* 5 (2017), pp. 135–146.
- [Cha+23] Nilesch Chakraborty, Jianqiang Li, Victor C. M. Leung, Samrat Mondal, Yi Pan, Chengwen Luo, and Mithun Mukherjee. “Honeyword-based Authentication Techniques for Protecting Passwords: A Survey”. In: *ACM Computing Surveys* 55.8 (2023), 169:1–169:37. DOI: [10.1145/3552431](https://doi.org/10.1145/3552431).
- [DMS24] Jimmy Dani, Brandon McCulloh, and Nitesh Saxena. *When AI Defeats Password Deception! A Deep Learning Framework to Distinguish Passwords and Honeywords*. 2024. arXiv: [2407.16964](https://arxiv.org/abs/2407.16964) [cs.CR].
- [DVA21] Antreas Dionysiou, Vassilis Vassiliades, and Elias Athanasopoulos. “HoneyGen: Generating Honeywords Using Representation Learning”. In: *Proceedings of the 2021 ACM Asia Conference on Computer and Communications Security (ASIA CCS)*. ACM, 2021, pp. 265–279. DOI: [10.1145/3433210.3453092](https://doi.org/10.1145/3433210.3453092).
- [FYM20] Muhammad Ali Fauzi, Bian Yang, and Edlira Martiri. “PassGAN for Honeywords: Evaluating the Defender and the Attacker Strategies”. In: *Advances on Smart and Soft Computing (ICACIn 2020)*. Vol. 1188. Advances in Intelligent Systems and Computing. Springer, 2020, pp. 391–401. DOI: [10.1007/978-981-15-6048-4_34](https://doi.org/10.1007/978-981-15-6048-4_34).
- [Goo+14] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. “Generative Adversarial Nets”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2014.
- [Hit+19] Briland Hitaj, Paolo Gasti, Giuseppe Ateniese, and Fernando Perez-Cruz. “PassGAN: A Deep Learning Approach for Password Guessing”. In: *Applied Cryptography and Network Security (ACNS)*. Vol. 11464. Lecture Notes in Computer Science. Springer, 2019, pp. 217–237. DOI: [10.1007/978-3-030-21568-2_11](https://doi.org/10.1007/978-3-030-21568-2_11).

- [JR13] Ari Juels and Ronald L. Rivest. “Honeywords: Making Password-Cracking Detectable”. In: *Proceedings of the 2013 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2013, pp. 145–160. DOI: [10.1145/2508859.2516671](https://doi.org/10.1145/2508859.2516671).
- [Kim14] Yoon Kim. “Convolutional Neural Networks for Sentence Classification”. In: *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2014, pp. 1746–1751.
- [Mel+16] William Melicher, Blase Ur, Sean M. Segreti, Saranga Komanduri, Lujo Bauer, Nicolas Christin, and Lorrie Faith Cranor. “Fast, Lean, and Accurate: Modeling Password Guessability Using Neural Networks”. In: *25th USENIX Security Symposium*. 2016, pp. 175–191.
- [Mik+13] Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg Corrado, and Jeffrey Dean. “Distributed Representations of Words and Phrases and their Compositionality”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2013.
- [NS05] Arvind Narayanan and Vitaly Shmatikov. “Fast Dictionary Attacks on Passwords Using Time-Space Tradeoff”. In: *Proceedings of the 12th ACM Conference on Computer and Communications Security (CCS)*. 2005, pp. 364–372.
- [Wei+09] Matt Weir, Sudhir Aggarwal, Breno de Medeiros, and Bill Glodek. “Password Cracking Using Probabilistic Context-Free Grammars”. In: *IEEE Symposium on Security and Privacy (S&P)*. 2009, pp. 391–405.
- [YV23] Fangyi Yu and Miguel Vargas Martin. “Honey, I Chunked the Passwords: Generating Semantic Honeywords Resistant to Targeted Attacks Using Pre-trained Language Models”. In: *Detection of Intrusions and Malware, and Vulnerability Assessment (DIMVA)*. Lecture Notes in Computer Science. Springer, 2023. DOI: [10.1007/978-3-031-35504-2_5](https://doi.org/10.1007/978-3-031-35504-2_5).
- [ZZL15] Xiang Zhang, Junbo Zhao, and Yann LeCun. “Character-Level Convolutional Networks for Text Classification”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2015.